

6CS012

Artificial Intelligence and Machine Learning

Portfolio Task 1: QnA

Full Name	:	Narayan Lohani
Student Number	:	2408869
Course	:	BSc (Hons) Computer Science
University Email	:	n.lohani@wlv.ac.uk
Group	:	L6CG15
Tutor	:	Jinu Nyachhyon
Module Leader	:	Siman Giri
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Long Question Answer

Answer for question no 1:: Identification of problems to leverage unsupervised learning for businesses

Problem Identification

Two high-impact problems include anomaly detection for fraud prevention and behavioral customer segmentation. These leverage unsupervised methods to process large-scale data, reducing fraud losses and enabling personalized services.

Fraud Anomaly Detection

In a real life context at digital payment systems, fraud detection involves scanning millions of daily transactions such as transfers, bill payments, or wallet loads. While scanning if there exists unusual patterns such as sudden high-value transfers to new recipients or atypical device location combinations that may be potential fraud attempts.

We can use Isolation Forest, an anomaly detection algorithm that isolates outliers by randomly partitioning data. This is ideal for large-scale, high-dimensional transaction logs as Isolation Forest has linear time complexity and minimal hyperparameters which is suitable for unlabeled data by assuming anomalies require fewer splits to isolate.

Isolation Forest can be deployed in real-time systems via streaming pipelines such as Apache Kafka. When the scores go above a threshold, it alerts for transaction holds or user verification. New thresholds are adapted on a fixed period (usually at night when traffic is low) by retraining on a new batch of aggregated data.

The most challenging part of using Isolation Forest for Fraud Detect is the model will get worse at detecting frauds when the fraudsters change their scamming method. To fix this, we have to regularly update the AI using recent data, even if we don't have the final 'answers' yet on which new transactions were definitely scams.

Customer Behavioral Segmentation

One way to tailor offers like cashback or payment limits is by profiling users based on unlabeled transaction data. This is done by distinguishing high-frequency small payments (daily expenses) and low-frequency large payments (bulk transactions).

Therefore, we can apply DBSCAN clustering, which groups variable-sized clusters based on density without predefined cluster counts. DBSCAN clustering performs very well for large-scale heterogeneous data by handling noise and outliers effectively, unlike K-Means.

To implement DBSCAN clustering, every week a big update is run on past week data which then categorizes users to find “High-Spender”, “Frequent Small Shopper”, “Potential Risk” types of users. These data are sent to marketing systems to send targeted ads, or to set spending limits for users who look risky.

The biggest challenge of DBSCAN is avoiding the curse of dimensionality. The raw data contains high-dimensional features like transaction timestamps and amounts. PCA, Latent Space of AutoEncoder can be used to reduce the dimensions.

Short Question Answer

Overfitting

Answer for question no 1 :: Explanation of Overfitting and Underfitting

Underfitting occurs when a model fails to learn the training data and does not capture the underlying patterns yielding high train error and high-test error as the model does not have enough complexity.

Overfitting occurs when a model learns the training data so precisely that it even learns the noise yielding low train error but high-test error because model cannot generalize correctly.

Both result in poor generalization performance. Underfitting cannot capture the patterns of the train data resulting in both high train loss and test loss (high bias) and Overfitting memorizes noise, causing a high difference between train loss and test loss (high variance). As neither has a balance between bias and variance, generalization is poor.

An example of underfitting can be a linear model fitting quadratic data. A linear model does not have enough complexity to capture the curvature pattern of the training data as well as test data.

An example of overfitting can be a high-degree polynomial regression model learning the noisy training data perfectly but failing to converge on test data as it memorized the fluctuations rather than trend of the data.

Neural Network Architecture

Answer for question no 2 :: FCN vs. Autoencoder

	Feedforward Neural Network	Autoencoder
Architecture	Input is fully connected to the hidden layer. Every hidden layer is connected to every forward hidden layer. The penultimate (last hidden) layer is connected to the output layer.	Multiple Convolutional layers with decreasing feature map size (Encoder) are connected compressing to bottleneck. Multiple Transposed Convolutional layers with increasing feature map (Decoder). It is expected to be symmetric with latent space in the middle.
Objective	Supervised learning for classification or regression (e.g., minimize cross-entropy loss).	Unsupervised reconstruction of input (e.g., minimize MSE between input/output).
Input-Output relationships	Input features + labels for outputs predictions.	Input only (self-supervised) for outputs reconstructed input.

For an image of handwritten digit “0” in Mnist dataset, an autoencoder will be used to reconstruct a noisy image of “0” to clean image of “0”, but a standard neural network will be used to classify the image as "0".

An application of autoencoders is anomaly detection (e.g., fraud detection in credit card transactions). The latent space of autoencoders forces dimensionality reduction which captures normal patterns in the latent space. However, upon reconstruction, error is low for normal transactions but high for anomalies.